

Chapter 4: Switch Realization

Section 4.1: Switch Applications

Erickson & Maksimović — Fundamentals of Power Electronics

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4.0 Background: Semiconductor Device Classes

Before analyzing switch operating points, it is essential to understand the two fundamental classes of power semiconductor devices.

Majority-Carrier Devices

Examples: Power MOSFET, Schottky diode.

Conduction occurs through the *majority carriers* of the doped region: electrons in N-type material, holes in P-type material. No minority carriers are injected across junctions.

Consequence for switching: When the device turns off, there are no stored minority carriers to remove. Turn-off is limited only by capacitive charging, making these devices fast.

Advantage	Disadvantage
Very fast switching (nanosecond range)	$R_{DS(on)}$ increases steeply with voltage rating
Low switching losses	Less efficient at high voltage than minority-carrier devices

Minority-Carrier Devices

Examples: BJT, IGBT, standard PN-junction diode, SCR, GTO.

During conduction, minority carriers are injected into the drift region. This *conductivity modulation* greatly reduces on-state resistance, enabling low conduction loss even at high voltages.

Consequence for switching: When the device turns off, injected minority carriers must be removed before the device can block voltage. This takes measurable time: **reverse recovery** in diodes, **storage time** in BJTs.

Advantage	Disadvantage
High breakdown voltage	Slow switching due to stored minority charge
Low on-state voltage drop	Reverse recovery causes current spikes and additional loss

The reverse recovery charge Q_{rr} of a freewheeling diode is deposited into the main transistor at every turn-on transition, creating a current spike proportional to $Q_{rr} \cdot f_{sw}$. This is often the dominant switching loss in hard-switched converters, and is the main reason fast-recovery or Schottky diodes are preferred in high-frequency designs.

Wide-Bandgap Semiconductors: SiC and GaN

Silicon carbide (SiC) and gallium nitride (GaN) devices use majority-carrier conduction but have a larger bandgap than silicon. This enables breakdown voltages comparable to minority-carrier silicon devices while maintaining the switching speed of silicon MOSFETs.

SiC and GaN break the fundamental silicon tradeoff between switching speed and voltage rating. They are the enabling technology for high-frequency, high-voltage converters in electric vehicles, solar inverters, and data center power supplies.

4.0.2 Switching Loss and the Clamped Inductive Load

Physical Mechanisms of Switching Loss

When a transistor switches a real inductive load, power is dissipated during every switching transition. Several mechanisms contribute:

1. **Voltage–current overlap:** during turn-off, v_{DS} rises to the bus voltage while i_D has not yet fallen to zero. Instantaneous power $p(t) = v_{DS}(t) \cdot i_D(t) \gg 0$ during this interval.
2. **Diode reverse recovery:** at transistor turn-on, the freewheeling diode's stored charge Q_{rr} must be swept out, causing an additional current spike through the transistor.
3. **Parasitic capacitance:** energy $\frac{1}{2}C_{DS}V^2$ stored in the transistor output capacitance is dissipated each switching cycle.
4. **Parasitic ringing:** oscillation between stray inductances and capacitances decays resistively before the end of the switching period.

Switching loss scales linearly with switching frequency:

$$P_{sw} \propto f_{sw}$$

Higher f_{sw} reduces magnetic component size (desirable) but increases switching loss (undesirable). Switching loss sets a practical upper limit on the operating frequency of hard-switched converters.

The Clamped Inductive Load

The *clamped inductive load* is the standard circuit model for characterizing transistor switching loss. It represents the switching cell of a DC-DC converter: a transistor in series with an inductive load, protected by a clamp.

The load contains an inductor L . By Faraday's law:

$$v_L(t) = L \frac{di_L(t)}{dt}$$

Inductor current cannot change instantaneously. If a circuit interrupts this current abruptly, the required voltage diverges and will destroy the transistor.

A clamp element (diode, Zener, or active circuit) limits the resulting voltage spike to a safe level. The most common implementation is a freewheeling diode from the load node to the supply rail.

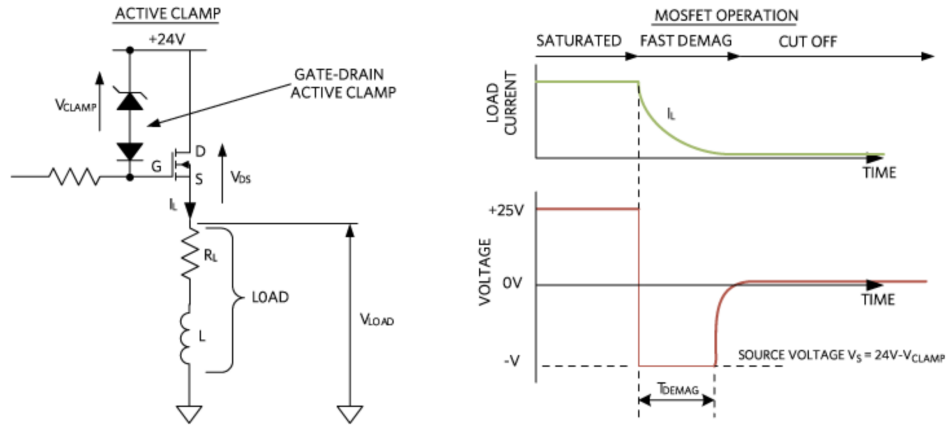


Figure 1: Active clamp circuit (clamped inductive load)

Operating sequence

1. **Transistor ON (saturated):** current flows $V_{cc} \rightarrow L \rightarrow T \rightarrow \text{GND}$. The transistor voltage $v_{DS} \approx 0$. The inductor stores energy.
2. **Transistor turns OFF:** the inductor maintains current. The load node voltage rises until the clamp diode becomes forward-biased and conducts. Current now re-circulates through $L \rightarrow D \rightarrow V_{cc}$.
3. **Fast demagnetization (T_{demag}):** the transistor drain is held at $v_{DS} = V_{cc}$ by the clamped node. Inductor current decays with time constant $\tau = L/R_L$.
4. **Cut-off:** inductor energy is fully dissipated. All currents reach zero.

Why v_{DS} is clamped at exactly V_{cc}

When the clamp diode conducts, the load node is connected to V_{cc} through the forward-biased diode ($V_f \approx 0$ for an ideal diode). The transistor's source is at ground. Therefore:

$$V_{\text{drain}} = V_{cc}, \quad V_{\text{source}} = 0 \quad \Rightarrow \quad V_{DS} = V_{cc}$$

If the node voltage attempts to exceed V_{cc} , the diode conducts harder and pulls it back. The bound is self-enforcing.

The clamp protects the transistor from overvoltage but does not eliminate switching loss. Reducing switching loss requires either lower f_{sw} , a faster device (smaller t_{fall}), or soft-switching techniques covered in later chapters.

4.1 Switch Applications

Every switch in a converter operating in steady state has a well-defined off-state voltage and on-state current. Plotting these on the i - v plane identifies which quadrant(s) the switch must cover. Device selection follows from matching the required quadrant(s) to the characteristics of available semiconductor devices.

4.1.1 Single-Quadrant Switches

The SPST Switch and Its Classification

The ideal SPST switch has two power terminals (0 and 1) with polarities. In the on-state, $v = 0$; in the off-state, $i = 0$. An optional control terminal C carries the gate signal.

Switches are classified by their control mechanism:

Type	Control terminal C	State determined by	Examples
Passive	None	Applied $v(t)$, $i(t)$ waveforms	Diode
Active	Yes	External control signal	MOSFET, BJT, IGBT, GTO

A passive switch has no gate driver and self-commutates based on circuit conditions. This is the freewheeling diode's key practical advantage: no additional control logic is required.

Device i - v Characteristics

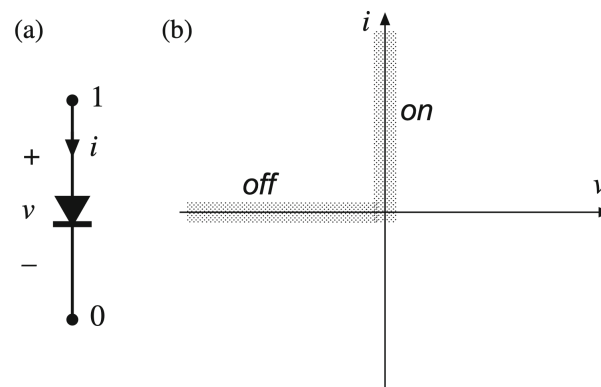
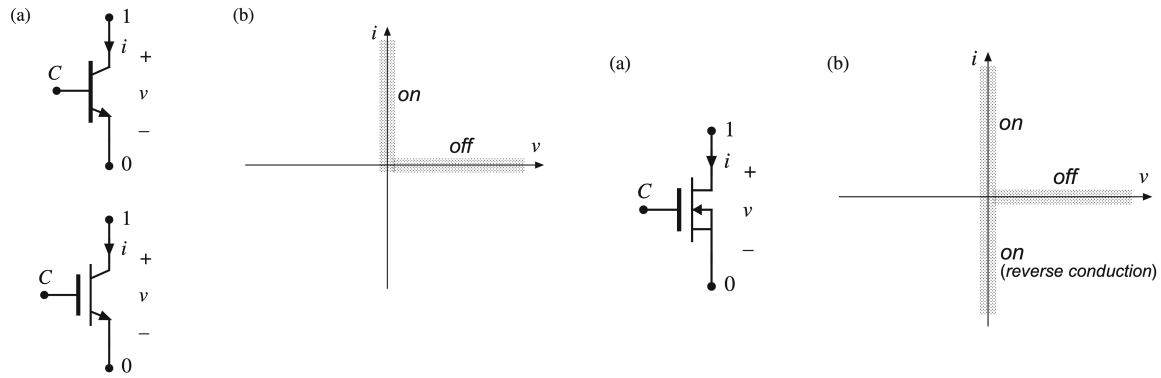


Figure 2: Idealized i - v characteristics: diode

Diode: blocks negative voltage ($v < 0$, $i = 0$); conducts positive current ($v = 0$, $i > 0$). Operates in Quadrant II.

BJT / IGBT: when OFF, blocks positive voltage ($v \geq 0$, $i = 0$); when ON, conducts positive current ($v = 0$, $i \geq 0$). Operates in Quadrant I. Reverse conduction and reverse blocking capabilities are poor or nonexistent.

Power MOSFET: same as BJT/IGBT in Quadrant I, but the intrinsic body diode permits conduction in the reverse direction (Quadrant III). Under normal operation, the MOSFET is used only in Quadrant I. The body diode is exploited in synchronous rectifier designs.

Figure 4: Idealized $i-v$ characteristics: PowerFigure 3: Idealized $i-v$ characteristics: MOSFET
BJT/IGBT

4.1.2 Current-Bidirectional Two-Quadrant Switches

Motivation: DC-AC Inverters

In a DC-AC inverter or servo amplifier, the output current alternates in sign over each AC cycle. The DC bus voltage is always positive. The switch must:

- Conduct **positive and negative** current $\Rightarrow$ Quadrants I and II
- Block only **positive** voltage

A single transistor or diode cannot satisfy both requirements simultaneously. A **current-bidirectional two-quadrant switch** is needed.

Implementation: Transistor with Antiparallel Diode

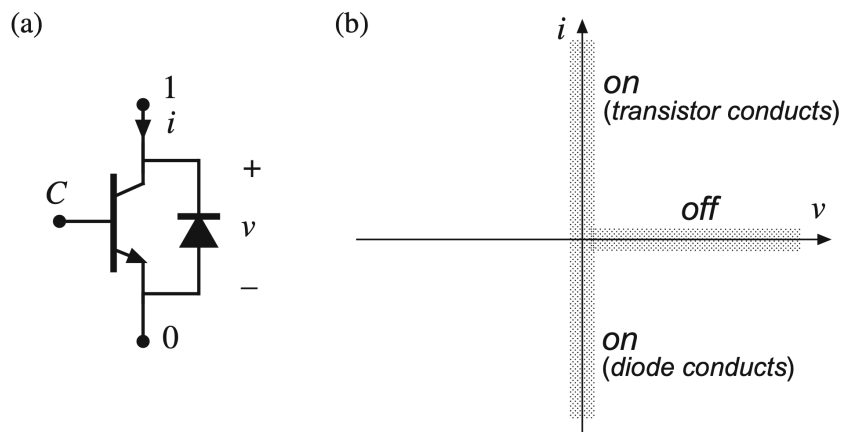


Figure 5: Current-bidirectional two-quadrant SPST switch

A transistor and diode are connected in **antiparallel** (reverse parallel):

- $i > 0$: transistor conducts in Quadrant I (controlled by gate signal)
- $i < 0$: antiparallel diode conducts in Quadrant II (self-commutated)
- Off-state ($v > 0$): transistor blocks; diode is reverse-biased

Single-Phase Inverter

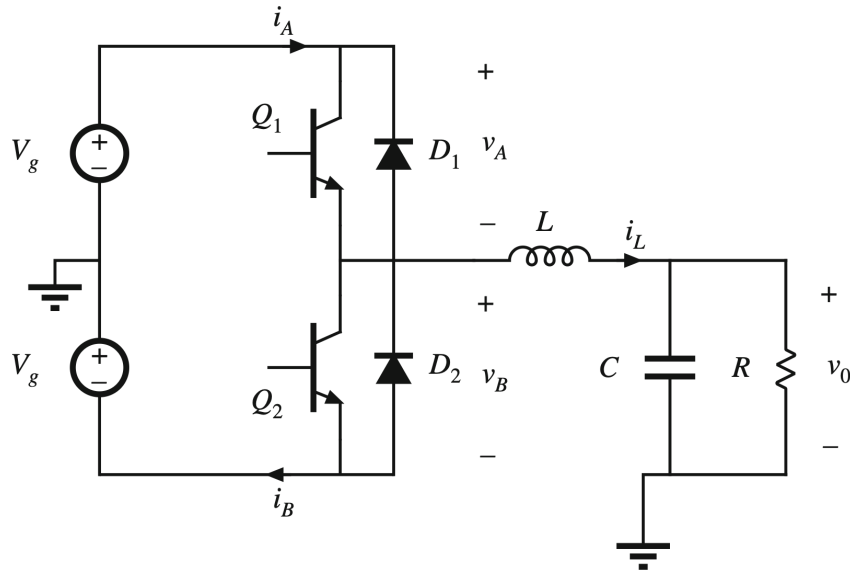


Figure 6: Single-phase DC-AC inverter using two current-bidirectional switches

Applying volt-second balance to the inverter inductor yields:

$$v_0 = (2D - 1) V_g \quad (1)$$

Duty cycle	Output voltage
$D = 1$	$v_0 = +V_g$
$D = 0.5$	$v_0 = 0$
$D = 0$	$v_0 = -V_g$

By varying the duty cycle sinusoidally as:

$$D(t) = 0.5 + D_m \sin(\omega t) \quad (2)$$

the output becomes sinusoidal and DC-to-AC inversion is achieved. The equilibrium inductor current is:

$$i_L = \frac{v_0}{R} = (2D - 1) \frac{V_g}{R} \quad (3)$$

At $D = 0.5$, $i_L = 0$. For $D > 0.5$, $i_L > 0$ and Q_1 , D_2 alternately conduct. For $D < 0.5$, $i_L < 0$ and Q_2 , D_1 alternately conduct. Neither a transistor alone nor a diode alone could handle both polarities of current — the antiparallel combination is essential.

Three-Phase Inverter

The voltage-source inverter (VSI) of Fig. 7 extends the same principle to three phases. Each phase requires one current-bidirectional SPDT switch, realized by two current-bidirectional SPST switches (IGBTs with antiparallel diodes). This topology is the standard inverter circuit in industrial variable-speed motor drives.

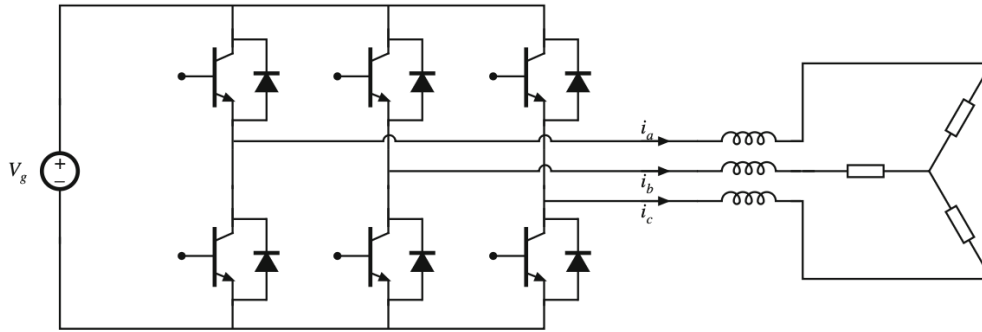


Figure 7: DC-to-3 ϕ ac voltage-source inverter (VSI). Three two-quadrant SPDT switches

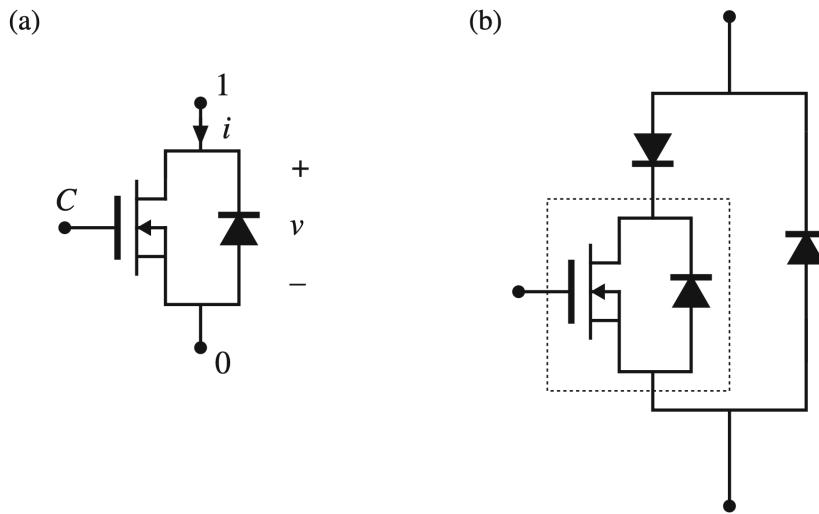


Figure 8: (a) Power MOSFET equivalent circuit showing the inherent body diode. (b) Addition of external series and antiparallel diodes to prevent body diode conduction

The Body Diode Problem in Power MOSFETs

Every power MOSFET contains an inherent **body diode** formed by the P-body and N-drain junction, connected antiparallel to the channel. Although this provides a reverse conduction path, using it is problematic:

- The body diode is a minority-carrier device with significant reverse recovery charge Q_{rr} .
- When the complementary MOSFET turns on, it must sweep Q_{rr} from the body diode, generating a large current spike.
- Some MOSFETs are not rated for these peak currents, leading to device failure.

The solution is to block the body diode with a series Schottky diode and add a separate fast-recovery antiparallel diode, as in Fig. 8(b). Modern power ICs often integrate MOSFETs with fast-recovery body diodes that make external diodes unnecessary.

4.1.3 Voltage-Bidirectional Two-Quadrant Switches

The application requires:

- Conduct only **positive** current
- Block **positive and negative** voltage $\Rightarrow$ Quadrants I and IV

A diode alone blocks only negative voltage; a transistor alone blocks only positive voltage. A series combination is required.

Implementation: Transistor in Series with Diode

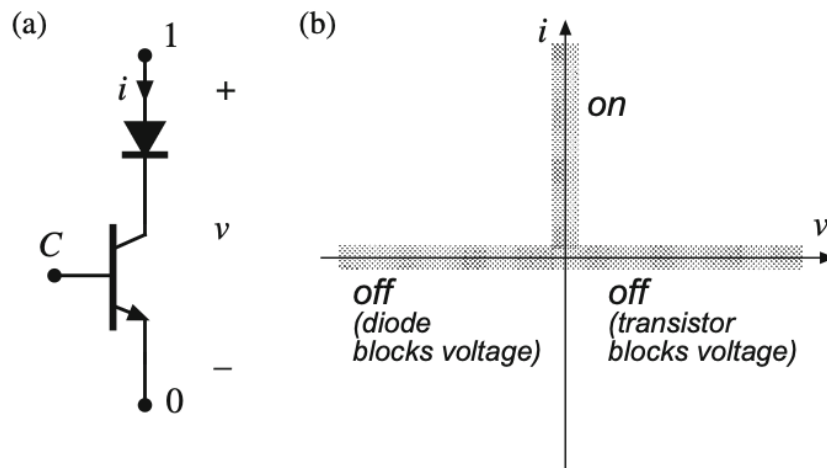


Figure 9: Voltage-bidirectional two-quadrant SPST switch

Key structural distinction between the two types of two-quadrant switch:

- **Antiparallel** diode $\Rightarrow$ adds a reverse current path $\Rightarrow$ current-bidirectional (Quadrants I+II)
- **Series** diode $\Rightarrow$ adds reverse voltage blocking $\Rightarrow$ voltage-bidirectional (Quadrants I+IV)

4.1.4 Four-Quadrant Switches

The most general case: both current and voltage can take either polarity. All four quadrants of the $i-v$ plane must be accessible.

Three Implementation Strategies

Method	Structure	Characteristic
(a)	Two current-bidirectional switches, back-to-back	Two gate drives required
(b)	Two voltage-bidirectional switches, antiparallel	Two gate drives required
(c)	Diode bridge + single transistor	Only one gate drive needed

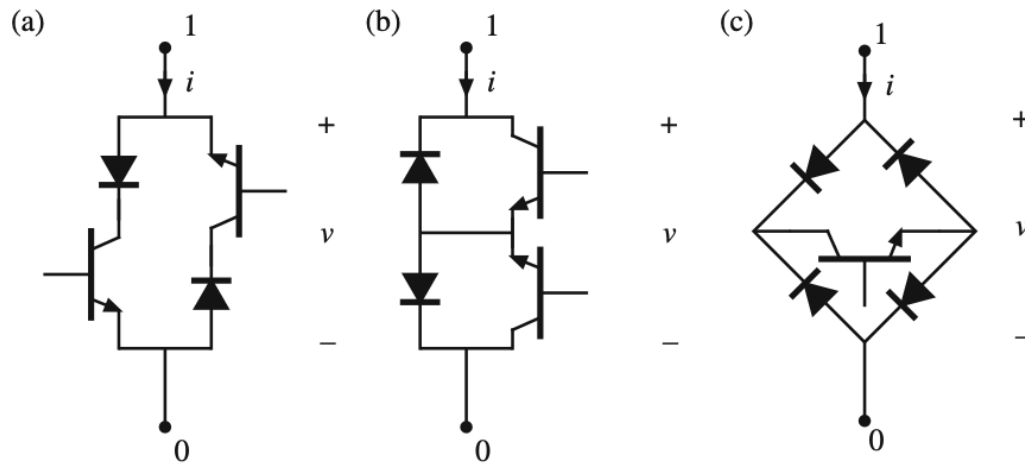


Figure 10: Three implementations of a four-quadrant SPST switch: (a) two current-bidirectional switches back-to-back; (b) two voltage-bidirectional switches in antiparallel; (c) diode bridge with a single controllable transistor

Summary of Section 4.1

Quadrant analysis: the universal device selection procedure

1. Decompose each SPDT switch into SPST switches.
2. For each SPST switch, determine off-state voltage polarity and on-state current polarity.
3. Map to the i - v plane. Identify the required quadrant(s).
4. Select the device or combination:

Application	Current	Voltage	Implementation
DC-DC	Unidirectional	Unidirectional	Transistor + freewheeling diode
DC-AC	Bidirectional	Unidirectional	Transistor + antiparallel diode
Bi DC-DC	Unidirectional	Bidirectional	Transistor + series diode
AC-AC	Bidirectional	Bidirectional	Four-quadrant combination